

Task Questions

1. Question 1

No, we cannot peer two VPCs that have the same CIDR ranges. This is because the IP addresses within each VPC must be unique across the peered VPCs to avoid any IP conflicts. If two VPCs have the same CIDR ranges, the IP addresses would not be unique, leading to routing issues like IP address conflicts and routing table confusion.

Since the same IP address could exist in both VPCs, it would be confusing where traffic should be directed when a request is made to that IP address. This will make the traffic to be sent to the wrong destination or not being delivered at all.

If two peered VPCs have the same CIDR range, the routing table wouldn't be able to determine the correct path for the traffic, leading to routing errors.

2. Question 2

After creating a peering connection, I need to add routes to the route tables of each VPC to direct traffic to the peered VPC. Specifically, I would add a route that points to the CIDR block of the peered VPC and targets the peering connection.

In the guided lab 7, I first configured the route table setting to the Lab VPC which is the source and set the destination traffic to the Shared VPC's IP range

Then, for the Shared VPC, I configure a route to direct traffic for the Lab VPC's IP range through the same peering connection.

3. Question 3

No, we cannot reach VPCC from VPCA directly even though they are both peered with VPCB. This is because AWS VPC peering connections are not transitive. Even though VPCA is peered with VPCB and VPCB is peered with VPCC, traffic cannot route from VPCA to VPCC through VPCB. To allow VPCA to communicate with VPCC, we would need to establish a separate peering connection between VPCA and VPCC.

Quiz

1. Knowledge Check for Module 7

This knowledge check was comprised of the most important questions throughout the module 7. The questions asked was kind of a summary to the whole module 7. In this knowledge check I was able to see my understanding of this module, and the from the mistakes I was able to correct myself of the parts that I was not sure about.

This knowledge check was mainly about connecting networks of VPCs from different to same account and even from different regions. The first question was what is AWS site-to-site VPN. Most of the questions were about the different techniques and ways available to connect to a different network securely, such as AWS direct connect, VPC peering, VPC endpoints, AWS Transit Gateway. There were scenario-based questions as well.

The screenshot shows the AWS Academy interface for the 'Module 7 Knowledge Check'. The left sidebar contains navigation links: Account, Dashboard, Courses, Calendar, Inbox, History, and Help. The main content area displays the following information:

- Breadcrumbs: ACav2EN-LTI13-75032 > Assignments > Module 7 Knowledge Check
- Section Header: Module 7 Knowledge Check
- Metadata: Due No Due Date, Points 100, Submitting an external tool
- Section Header: Module 7 knowledge check results
- Score Summary:

Your score:	70%	(70 points)
Required score:	70%	(70 points)
- Result: Congratulations! You have completed this module.
To continue, choose **Next** in the lower-right corner.

Module 7 - Guided Lab: Creating a VPC Peering Connection

Task 1: Creating a VPC peering connection

I'm setting up a VPC peering connection between two VPCs, "Lab VPC" and "Shared VPC". This allows them to communicate privately. I create the connection, name it "Lab-Peer", and specify the requester (Lab VPC) and accepter (Shared VPC).

Afterward, I accept the connection request for Shared VPC, completing the setup.

The screenshot shows the AWS VPC dashboard with a green notification banner at the top stating: "A VPC peering connection pcx-0fd566c9295ac5552 / Lab-Peer has been requested." The main content area displays the details for the peering connection "pcx-0fd566c9295ac5552 / Lab-Peer". A "Pending acceptance" message indicates that the requester has sent a request and the accepter must accept it by Saturday, May 18, 2024, at 15:29:36 GMT+5:30. The details table shows the requester owner ID (340250290150), requester VPC (vpc-03f80f9d39509dbb3 / Lab VPC), and requester region (N. Virginia (us-east-1)). The accepter owner ID is also 340250290150, and the accepter VPC is vpc-08bd9e65ac1b259f4 / Shared VPC, also in N. Virginia (us-east-1). The status is "Pending Acceptance by 340250290150".

Details	
Requester owner ID	340250290150
Requester VPC	vpc-03f80f9d39509dbb3 / Lab VPC
Requester CIDRs	10.0.0.0/16
Requester Region	N. Virginia (us-east-1)
Accepter owner ID	340250290150
Accepter VPC	vpc-08bd9e65ac1b259f4 / Shared VPC
Accepter CIDRs	-
Accepter Region	N. Virginia (us-east-1)

The screenshot shows the AWS VPC dashboard with a green notification banner at the top stating: "Your VPC peering connection (pcx-0fd566c9295ac5552 / Lab-Peer) has been established." The main content area displays the "Peering connections (1)" list. The connection "Lab-Peer" is now listed with a status of "Active". The requester VPC is "vpc-03f80f9d39509dbb3 / Lab ..." and the accepter VPC is "vpc-08bd9e65ac1b259f4 / Shar...". A "Create peering connection" button is visible in the top right corner of the list.

Name	Peering connection ID	Status	Requester VPC	Accepter VPC
Lab-Peer	pcx-0fd566c9295ac5552	Active	vpc-03f80f9d39509dbb3 / Lab ...	vpc-08bd9e65ac1b259f4 / Shar...

Task 2: Configuring route tables

I'm configuring route tables for both VPCs to route traffic through the newly created peering connection.

For the "Lab VPC", I set up a route to send traffic destined for the Shared VPC's IP range (10.5.0.0/16) through the peering connection named "Lab-Peer".

VPC dashboard × Updated routes for rtb-0242a1fc3cc6c4f91 / Lab Public Route Table successfully

EC2 Global View Filter by VPC: Select a VPC

Virtual private cloud Your VPCs Subnets **Route tables** Internet gateways Egress-only internet gateways Carrier gateways DHCP option sets Elastic IPs Managed prefix lists Endpoints Endpoint services NAT gateways Peering connections

rtb-0242a1fc3cc6c4f91 / Lab Public Route Table Actions

Details Info

Route table ID rtb-0242a1fc3cc6c4f91	Main No	Explicit subnet associations subnet-07df546a5f9fcaa20 / Lab Public Subnet	Edge associations -
VPC vpc-03f80f9d39509dbb3 Lab VPC	Owner ID 340250290150		

Routes Subnet associations Edge associations Route propagation Tags

Routes (3) Both Edit routes

Destination	Target	Status	Propagated
0.0.0.0/0	igw-02d8e7c5deb5849c8	Active	No
10.0.0.0/16	local	Active	No
10.5.0.0/16	pcx-0fd566c9295ac5552	Active	No

Then, for the Shared VPC, I configure a route to direct traffic for the Lab VPC's IP range (10.0.0.0/16) through the same peering connection.

These configurations ensure seamless communication between the two VPCs via the peering connection.

VPC dashboard × Updated routes for rtb-008d9f495140ab632 / Shared-VPC Route Table successfully

EC2 Global View Filter by VPC: Select a VPC

Virtual private cloud Your VPCs Subnets **Route tables** Internet gateways Egress-only internet gateways Carrier gateways DHCP option sets Elastic IPs Managed prefix lists Endpoints Endpoint services NAT gateways Peering connections

rtb-008d9f495140ab632 / Shared-VPC Route Table Actions

Details Info

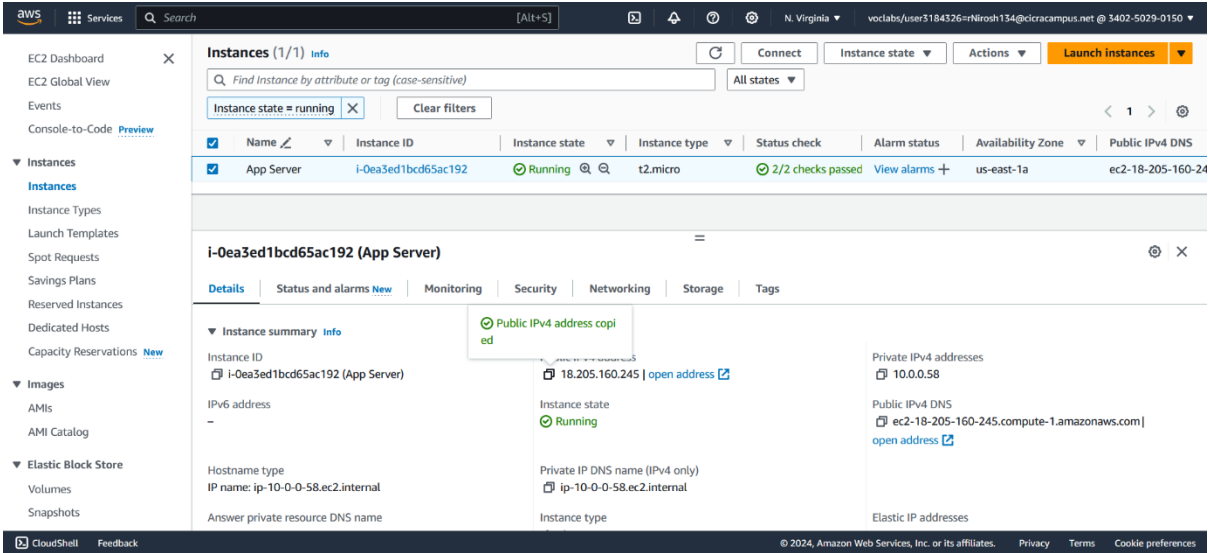
Route table ID rtb-008d9f495140ab632	Main No	Explicit subnet associations 2 subnets	Edge associations -
VPC vpc-08bd9e65ac1b259f4 Shared VPC	Owner ID 340250290150		

Routes Subnet associations Edge associations Route propagation Tags

Routes (2) Both Edit routes

Destination	Target	Status	Propagated
10.0.0.0/16	pcx-0fd566c9295ac5552	Active	No
10.5.0.0/16	local	Active	No

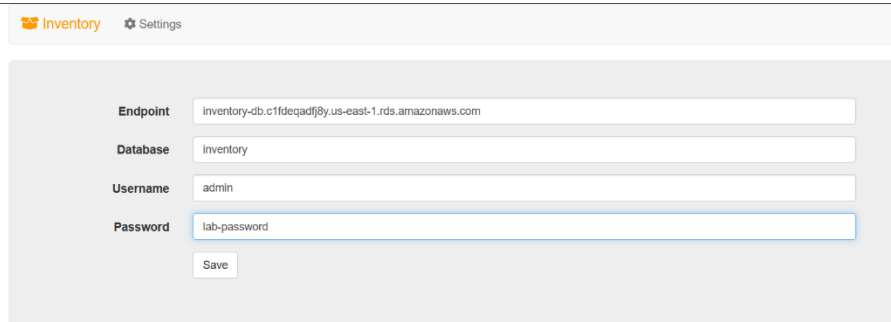
I'm testing the VPC peering connection I set up earlier.



I copy the IPv4 Public IP address of the Inventory application from the EC2 instance.



Then, I paste it into a new browser tab and configure the settings to connect to the database: Endpoint (copied from details), Database ("inventory"), Username ("admin"), and Password ("lab-password").



Inventory Settings

Endpoint:

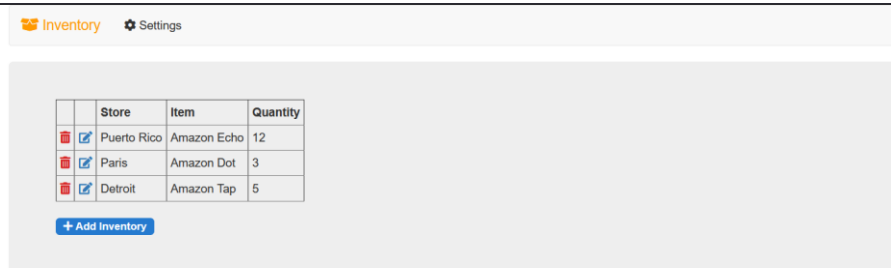
Database:

Username:

Password:

Save

After saving, the application should display data from the database, confirming the VPC peering connection's success. Since the Shared VPC lacks an internet gateway, this confirms that the only way to access the database is through the VPC peering connection.



Inventory Settings

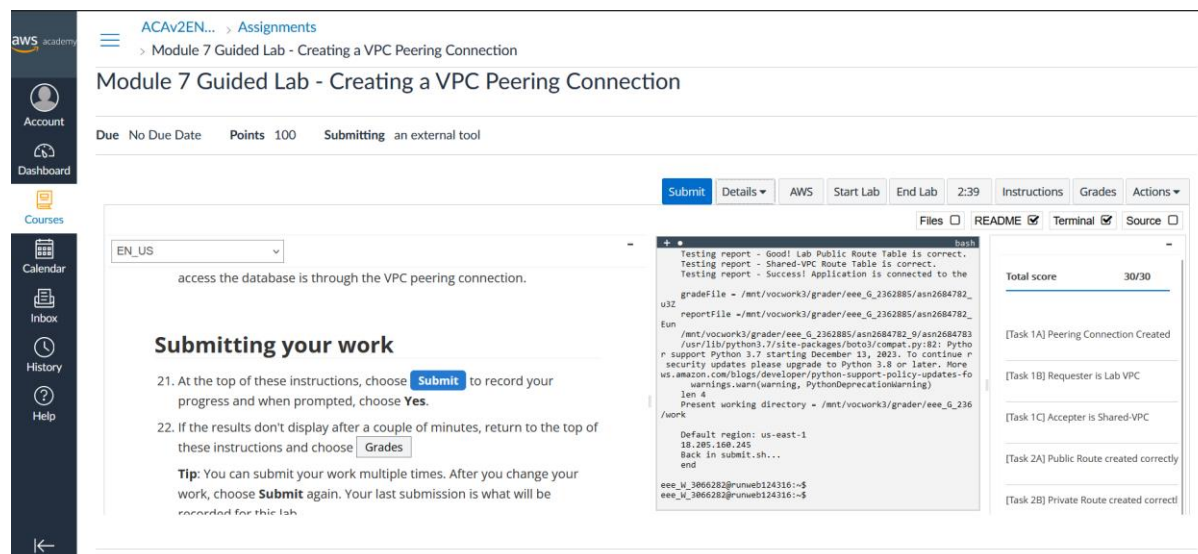
	Store	Item	Quantity
☒	Puerto Rico	Amazon Echo	12
☒	Paris	Amazon Dot	3
☒	Detroit	Amazon Tap	5

+ Add Inventory

This page was generated by instance i-0ea3ed1bcd65ac192 in Availability Zone us-east-1a.

Grade:

My grade is 30/30



ACAv2EN... > Assignments

> Module 7 Guided Lab - Creating a VPC Peering Connection

Module 7 Guided Lab - Creating a VPC Peering Connection

Due No Due Date Points 100 Submitting an external tool

Submit Details AWS Start Lab End Lab 2:39 Instructions Grades Actions

Files README Terminal Source

EN_US

access the database is through the VPC peering connection.

Submitting your work

21. At the top of these instructions, choose **Submit** to record your progress and when prompted, choose **Yes**.

22. If the results don't display after a couple of minutes, return to the top of these instructions and choose **Grades**.

Tip: You can submit your work multiple times. After you change your work, choose **Submit** again. Your last submission is what will be recorded for this task.

```
Testing report - Good! Lab Public Route Table is correct.
Testing report - Shared-VPC Route Table is correct.
Testing report - Success! Application is connected to the
gradeFile = /mnt/vocwork3/grader/eee_g_2362885/asn2684782_
reportFile = /mnt/vocwork3/grader/eee_g_2362885/asn2684782_
Eun /mnt/vocwork3/grader/eee_g_2362885/asn2684782_9/asn2684783
/usr/lib/python3.7/site-packages/boto3/compat.py:82: Pytho
r support Python 3.7 starting December 13, 2023. To continue r
security updates please upgrade to Python 3.8 or later. More
vs.amazon.com/blogs/developer/python-support-policy-updates-fo
warnings.warn(warning, PythonDeprecationWarning)
len 4
Present working directory = /mnt/vocwork3/grader/eee_g_236
/work
Default region: us-east-1
$B_JMS.isb.245
Back in submit.sh...
end
eee_W_3866282@runweb124316:~$
eee_W_3866282@runweb124316:~$
```

Total score 30/30

[Task 1A] Peering Connection Created

[Task 1B] Requester is Lab VPC

[Task 1C] Accepter is Shared-VPC

[Task 2A] Public Route created correctly

[Task 2B] Private Route created correctly